BODY—COLLISION REPAIR UPDATES	Article No. 02-17-11

FORD: 2003 CROWN VICTORIA

LINCOLN: 2003 TOWN CAR

MERCURY: 2003 GRAND MARQUIS

ISSUE

The 2003 Crown Victoria, Grand Marquis, and Town Car have been redesigned to improve crashworthiness and damage resistance. These changes will affect current collision repair processes.

ACTION

The purpose of this article is to summarize the key changes, diagnostic steps to consider during collision repairs, and Ford-recommended collision repair procedures.

SERVICE INFORMATION

The 2003 Crown Victoria, Grand Marquis, and Town Car have been significantly redesigned. Most of these design revisions are not obvious. The purpose of this Technical Service Bulletin is to summarize the key changes, diagnostic steps to consider during collision repairs, and Ford-recommended collision repair procedures.

Aluminum Rocker Panel Reinforcement System (Figure 1)

The rocker panels have an extruded aluminum reinforcement added inside to augment crashworthiness. The purpose of the reinforcement is to reduce passenger compartment displacement during a front offset crash.

NOTE
THE ALUMINUM ROCKER PANEL REINFORCEMENT IS STRUCTURALLY BONDED AND BOLTED TO THE SERVICE FRONT FLOOR PAN ASSEMBLY.

This aluminum reinforcement has the following effect on collisions and repairs:

1. The vehicle will deform differently during front offset and side collisions. The aluminum reinforcement will resist deformation, and likely transfer collision impact energy to adjacent areas of the body.
2. Due to the increased stiffness, the rocker panel area will be more resistant to straightening operations than previous model year vehicles without the reinforcement.
3. The aluminum reinforcement is not repairable or serviceable. If there is evidence that the reinforcement is bent, or the mounting plane has been compromised (creased, dented, etc.) it must be replaced, along with the service part front floor pan assembly.
4. The service part front floor pan, with the aluminum reinforcement, cannot be used in prior model Crown Victorias, Grand Marquis, or Town Cars.

Chassis Revisions (Figure 2)

There have been four significant changes to the chassis of all three vehicles, to improve crashworthiness, NVH and repairability:

1. **Hydro Formed Front Section:** The front chassis rails, up to the torque box are Hydro Formed. In conjunction with the new bumper system (Figure 3), they should be capable of withstanding front barrier impacts of up to approximately 17 mph (27 km/h) without damage.
2. **Front Rail Sectioning Repair Kits:** For front collisions where damage has occurred to the front hydro formed areas, front rail sectioning kits can be purchased. The kits consist of a replacement RH or LH front frame sections and instruction sheets.
3. **High-Strength Steel Side Members:** The side rail material has been changed to high-strength steel. No heat is allowed on these parts during pulling operations, to stress-relieve the parts. If the welds have been broken between the side rails and the front torque boxes, use the recommended repair procedure included with this TSB to repair the welds.

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4. **Integral Structural Foam Inserts:** Structural foam inserts are located in four places on the chassis, ahead of the front torque boxes and above the rear axle kick up. Their purposes are to enhance the crashworthiness and NVH of the vehicles. These inserts consist of injection-molded plastic supporting structure, and structural foam, bonding the plastic carrier with the inside of the frame rails. Because of their locations in the chassis, they may not be serviceable or repairable.

Front Bumper System (Figure 4)

The front bumper system consists of the following new elements:

1. TPO bumper cover.
2. Injection-molded plastic energy absorbers, capable of absorbing more energy than the previous design, riveted to the beam.
3. Martensitic steel bumper, bolt-on beam. This part cannot be heated or welded.
4. Front bumper, bolt-on crush cans, increasing crush space for frontal impacts and capable of absorbing more energy than the previous design (up to approximately 17 mph (27 km/h)).

Extended Rear Park Assist (Town Car) (Figure 5)

Lincoln Town Car models will offer optional rear backup sensors. The center sensor uses radar technology, and is located behind the bumper cover, above the license plate mounting area.

1. If the mounting bracket is bent only, but the unit otherwise is functional, the bracket can be straightened out and the sensor re-used.
2. If more than the bracket is damaged, the complete sensor unit must be replaced.

Inner Door - Lincoln Town Car Only (Figure 6)

The Lincoln Town Car door for 2003 has had a hole design revision to the lower left inner door panel.

1. The change was from a 7.2 mm square hole to 10.0 mm round hole. If the 2003 level door is used on a previous model year Town Car this hole will have to be drilled out.
2. Apply Motorcraft TA-8 (Superseal) inside the hole after drilling.

PART NUMBER	PART NAME
TA-8	Motorcraft Superseal

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 111000, 305000

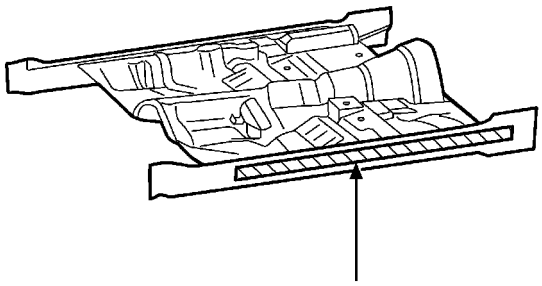


Figure 1 - Article 02-17-11

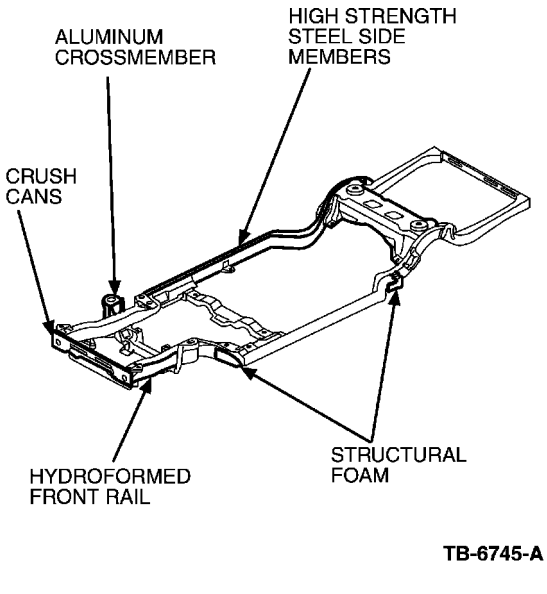


Figure 2 - Article 02-17-11

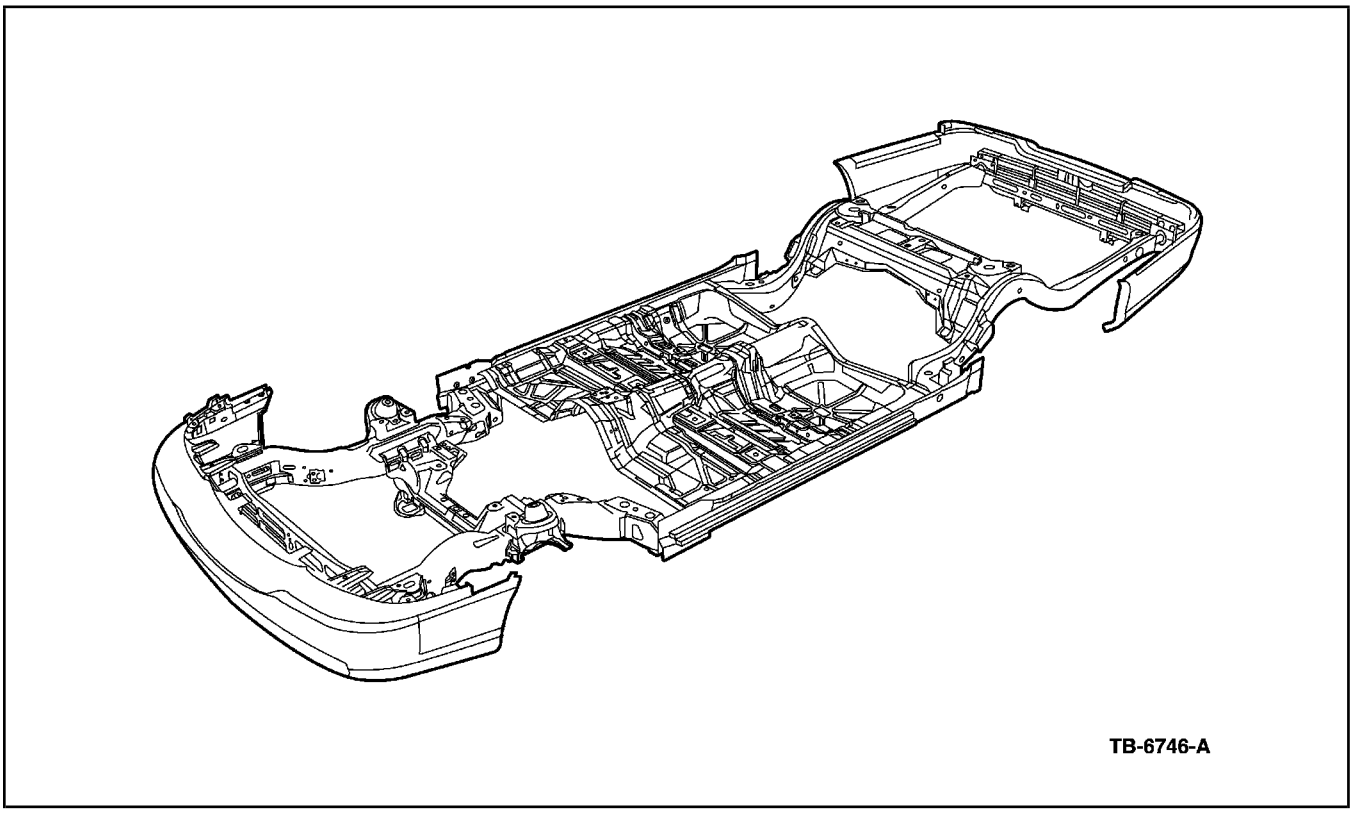


Figure 3 - Article 02-17-11

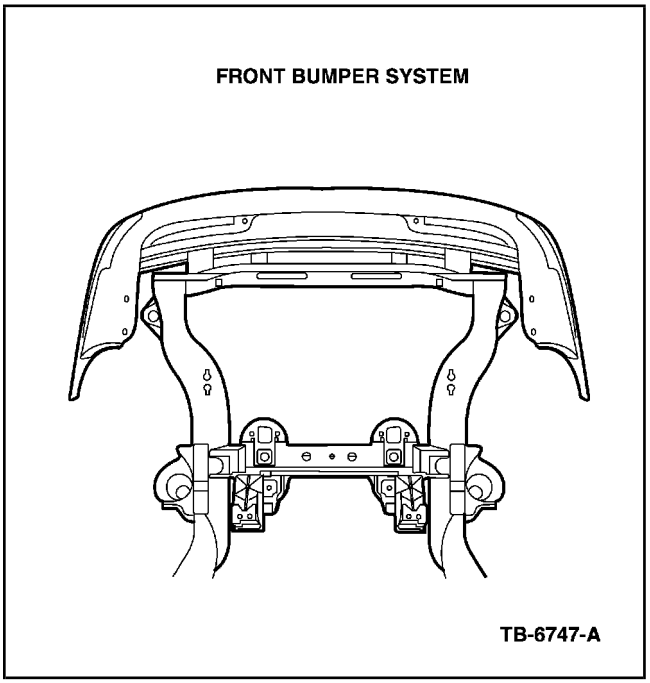


Figure 4 - Article 02-17-11

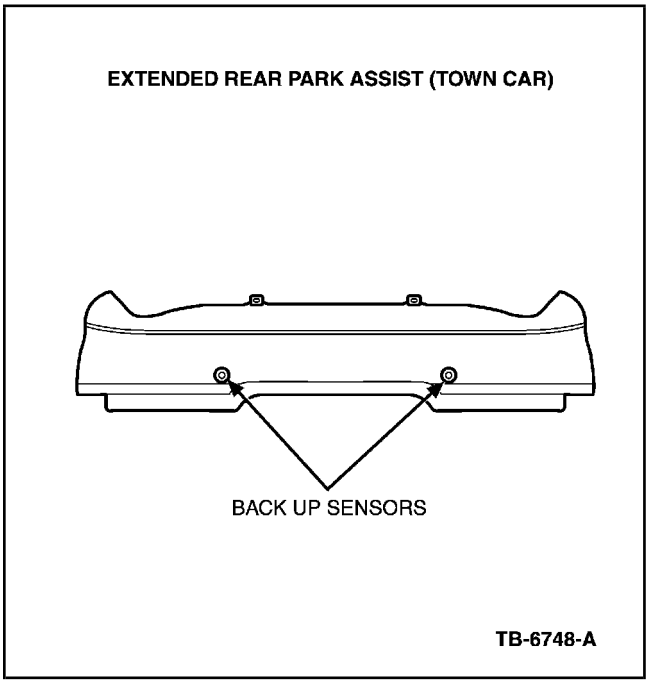


Figure 5 - Article 02-17-11

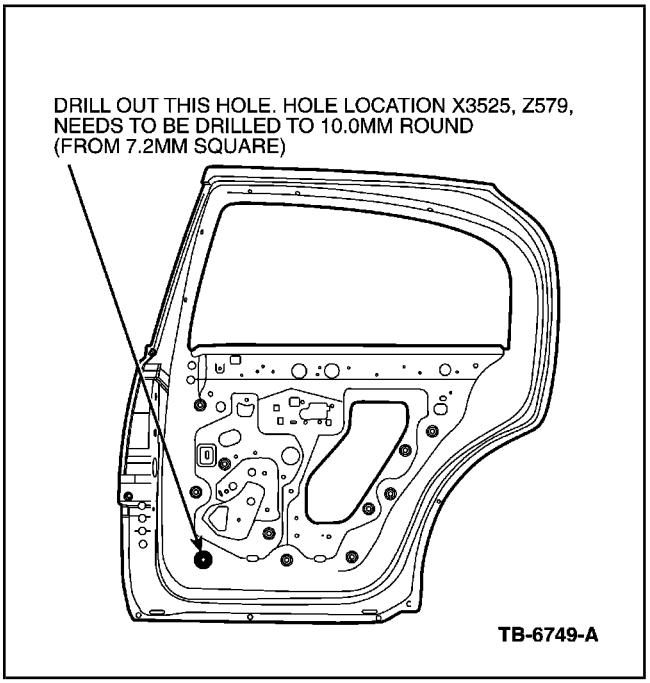


Figure 6 - Article 02-17-11

NOTE: The information in Technical Service Bulletins is intended for use by trained, professional technicians with the knowledge, tools, and equipment to do the job properly and safely. It informs these technicians of conditions that may occur on some vehicles, or provides information that could assist in proper vehicle service. The procedures should not be performed by "do-it-yourselfers". Do not assume that a condition described affects your car or truck. Contact a Ford, Lincoln, or Mercury dealership to determine whether the Bulletin applies to your vehicle.